

SAFETRIAGE: A SAFETY-GATED FRAMEWORK FOR CLINICAL AI DECISION-MAKING UNDER UNCERTAINTY

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A prototype clinical AI system that models when to answer, defer, or escalate under uncertainty in high-stakes clinical care.

◆ 1. Problem

Clinical AI systems are typically evaluated based on predictive accuracy. However, accuracy alone does not determine whether a system is safe to use in real-world care.

In clinical settings:

- Information is often incomplete
- Uncertainty is unavoidable
- Incorrect autonomous decisions can cause harm

The key challenge is not just prediction but ensuring **safe decision behavior under uncertainty**.

◆ 2. Approach

We developed a prototype system consisting of:

Baseline Model

A multiclass classifier predicting:

- Low
- Moderate
- Urgent
- Emergent

Safety-Gated Action Layer

A rule-based layer that determines whether to:

- **Answer** (provide output)

- **Defer** (insufficient confidence or missing information)
- **Escalate** (high-risk signals present)

Safety Triggers

- High-risk rule activation
- Missing critical inputs
- Low confidence
- Low agreement

◆ 3. Key Results

Metric	Value
Baseline Accuracy	0.68
Unsafe Raw Output Rate	0.0975
Unsafe Gated Output Rate	0.0000
Defer Rate	0.28
Escalate Rate	0.58
Answer Rate	0.1375

◆ 4. Interpretation

The safety-gated framework eliminated unsafe autonomous outputs while redistributing decisions toward deferral and escalation.

These results demonstrates that:

- Safe clinical AI is not defined solely by accuracy
- Decision control under uncertainty is essential
- Systems must be evaluated on **when they should not act**

This shift highlights that preventing unsafe action is as important as making correct predictions.

◆ 5. Key Insight

Clinical AI evaluation should move from:

Prediction quality → Decision behavior

Specifically:

- Not just “*Is the model correct?*”
- But “*Did the system act safely?*”

◆ 6. Limitations

- Based on simulated triage data
- Rule-based safety layer (not adaptive yet)
- No integration with real clinical workflows

Further validation is required in real-world settings.

◆ 7. Future Directions

- Prospective evaluation in clinical environments
- Adaptive safety gating with feedback
- Integration into clinical decision support systems
- Extension to multimodal inputs (text, imaging, signals)

◆ 8. Project Link

Code and Repository:

<https://github.com/erhaborAI/safe-triage-clinical-ai>